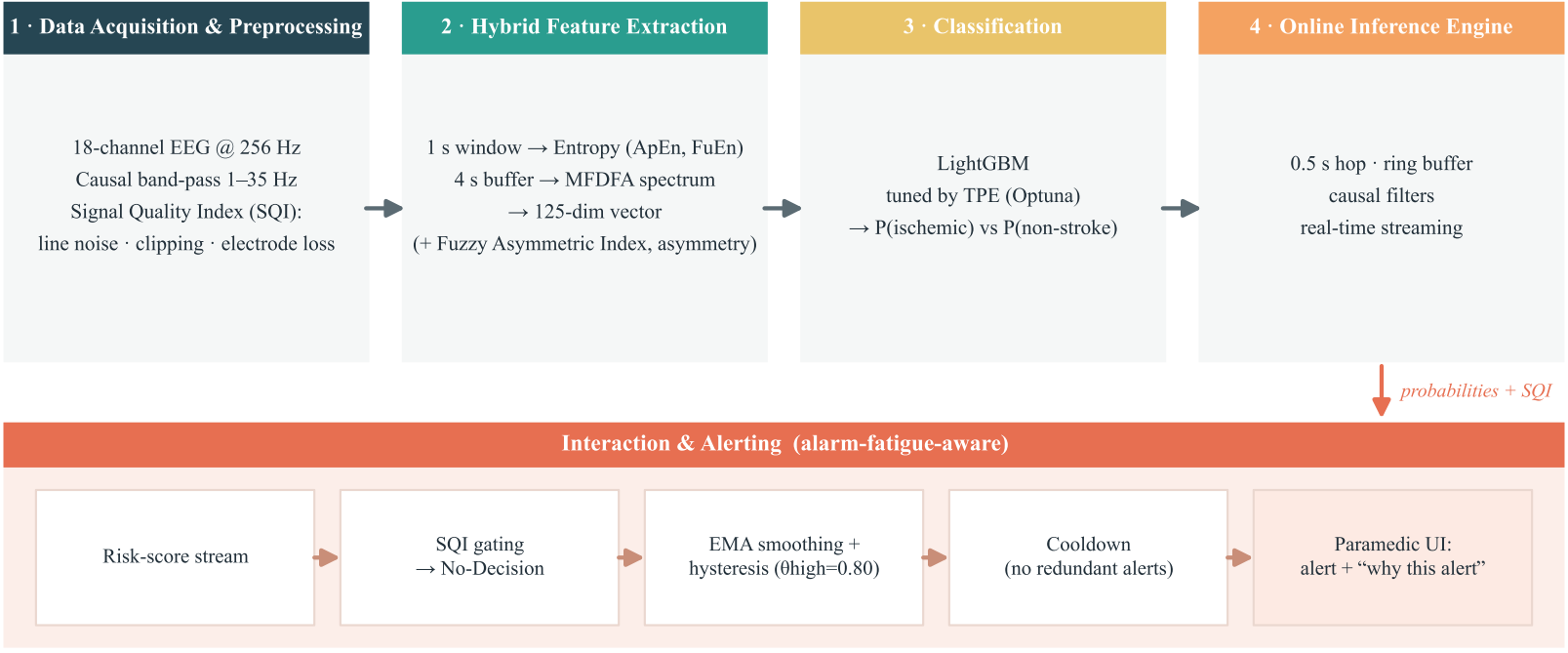

1 · Data Acquisition & Preprocessing
18-channel EEG @ 256 Hz
Causal band-pass 1–35 Hz
Signal Quality Index (SQI):
line noise · clipping · electrode loss
2 · Hybrid Feature Extraction
1 s window → Entropy (ApEn, FuEn)
4 s buffer → MFDFA spectrum
→ 125-dim vector
(+ Fuzzy Asymmetric Index, asymmetry)
3 · Classification
LightGBM
tuned by TPE (Optuna)
→ P(ischemic) vs P(non-stroke)
4 · Online Inference Engine
0.5 s hop · ring buffer
causal filters
real-time streaming
probabilities + SQI
Interaction & Alerting (alarm-fatigue-aware)
Risk-score stream
SQI gating
→ No-Decision
EMA smoothing +
hysteresis (θhigh=0.80)
Cooldown
(no redundant alerts)
Paramedic UI:
alert + “why this alert”